

Is the Netherlands on track?

CO₂ emissions per person in the Netherlands compared with the EU-27, 1990–2024

Why this matters

SDG 13 Climate Action

Target 13.2: integrate climate measures into national policy. Indicator 13.2.2: total greenhouse gas emissions per year. This dashboard looks at CO₂ per person instead, because it allows a fair comparison between countries of very different sizes.

The problem

In 2025 the world was 1.43°C warmer than before industrialisation (WMO, 2026). The EU and the Netherlands must cut emissions by 55% by 2030 compared with 1990 (European Climate Law; Dutch Climate Act).

But PBL (2026) gives the Netherlands less than a 5% chance of reaching its 2030 target.

Sources: WMO (2026); PBL (2026), Klimaat- en Energieverkenning 2026; European Climate Law (2021); Dutch Climate Act (Klimaatwet).

Research question

How have CO₂ emissions per capita in the Netherlands developed between 1990 and 2024 compared to the other EU-27 member states?

SQ1 How did Dutch emissions change, and when?

SQ2 Where does the Netherlands rank in the EU?

SQ3 Which countries reduced the most?

SQ4 Where is the Netherlands heading by 2030?

Per capita the average tonnes of CO₂ emitted per person per year inside a country (imports not counted).

Who is this for? Dutch government (Minister of Climate and Green Growth), policy analysts (PBL), energy companies and industry, provinces and municipalities, citizens and environmental organisations.

Data & Method

Where the data comes from, how it was cleaned and what was calculated.

1 Data

Source

Our World in Data – “CO₂ emissions per capita”, based on the Global Carbon Budget (2025).

Format

CSV file, 26,509 rows and 4 columns: Entity, Code, Year and CO₂ emissions per capita (tonnes per person per year).

Quality issues found

- Regions such as “World” and “Europe” mixed with countries
- Kosovo has an OWID code but is a country
- Countries start in different years
- Dutch settings misread the decimals and the “₂”
- No missing values and no duplicates

2 Preparation (Power Query)

26,509 → 8,068 → 980 rows

- 1 Import CSV (UTF-8, English locale) → 26,509
- 2 Rename columns (Country, CO₂_per_capita_tonnes)
- 3 Keep the years 1990–2024 → 8,068
- 4 Keep the 27 EU countries + EU average → 980
- 5 Rename to “EU-27 average”
- 6 Add a Type column (Country / EU-27 average)
- 7 Round CO₂ values to 3 decimals

Result

980 rows = 28 × 35 years. No errors, no empty values, no duplicates. The raw file stays unchanged, so Refresh repeats every step.

3 Calculations (DAX)

CO₂ per person

AVERAGE of the CO₂ values

CO₂ 1990 / CO₂ 2024

CALCULATE with a year filter

Change since 1990

DIVIDE(2024 – 1990, 1990)

NL rank 2024

Counts countries above the Netherlands + 1

Statistics

Standard deviation between EU countries (STDEV.P)
5.58 t (1990) → 1.56 t (2024)

Correlation Year – Dutch CO₂, 2010–2024
r = –0.95

Correlation 1990 level – % change
r = –0.65

Straight-line trend of Dutch CO₂ to 2030
5.0 t (trend from 2010) to 4.4 t (trend from 2014)

Design choices

Netherlands is always blue, the **EU-27 average** black and **other countries** grey. Line charts show trends over time, sorted bar charts compare countries and cards hold the key numbers. Hover over any bar or line for exact values.

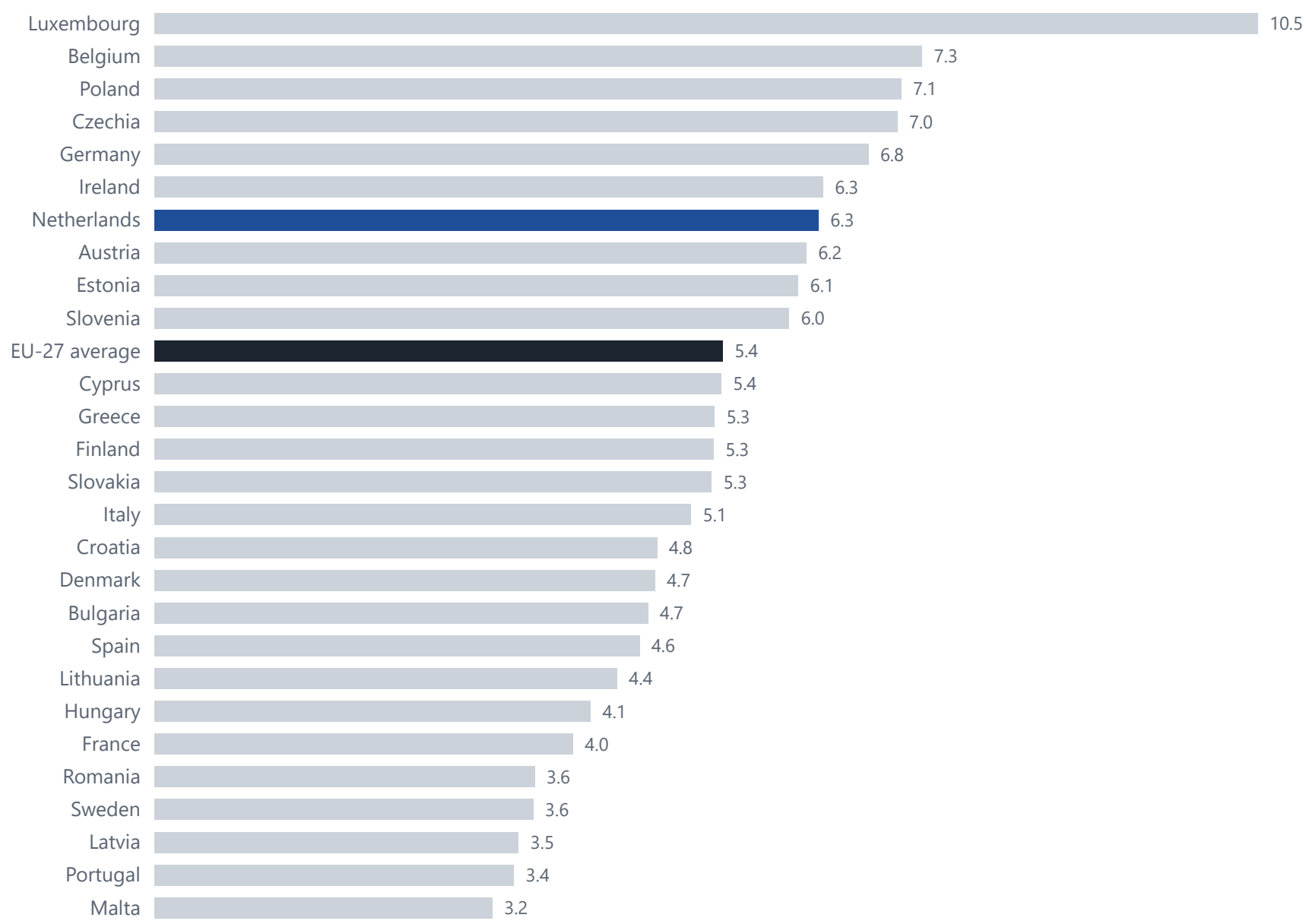
How the Netherlands compares

CO₂ emissions per person across the EU-27 from 1990 to 2024, in tonnes per person per year. Source: Our World in Data.



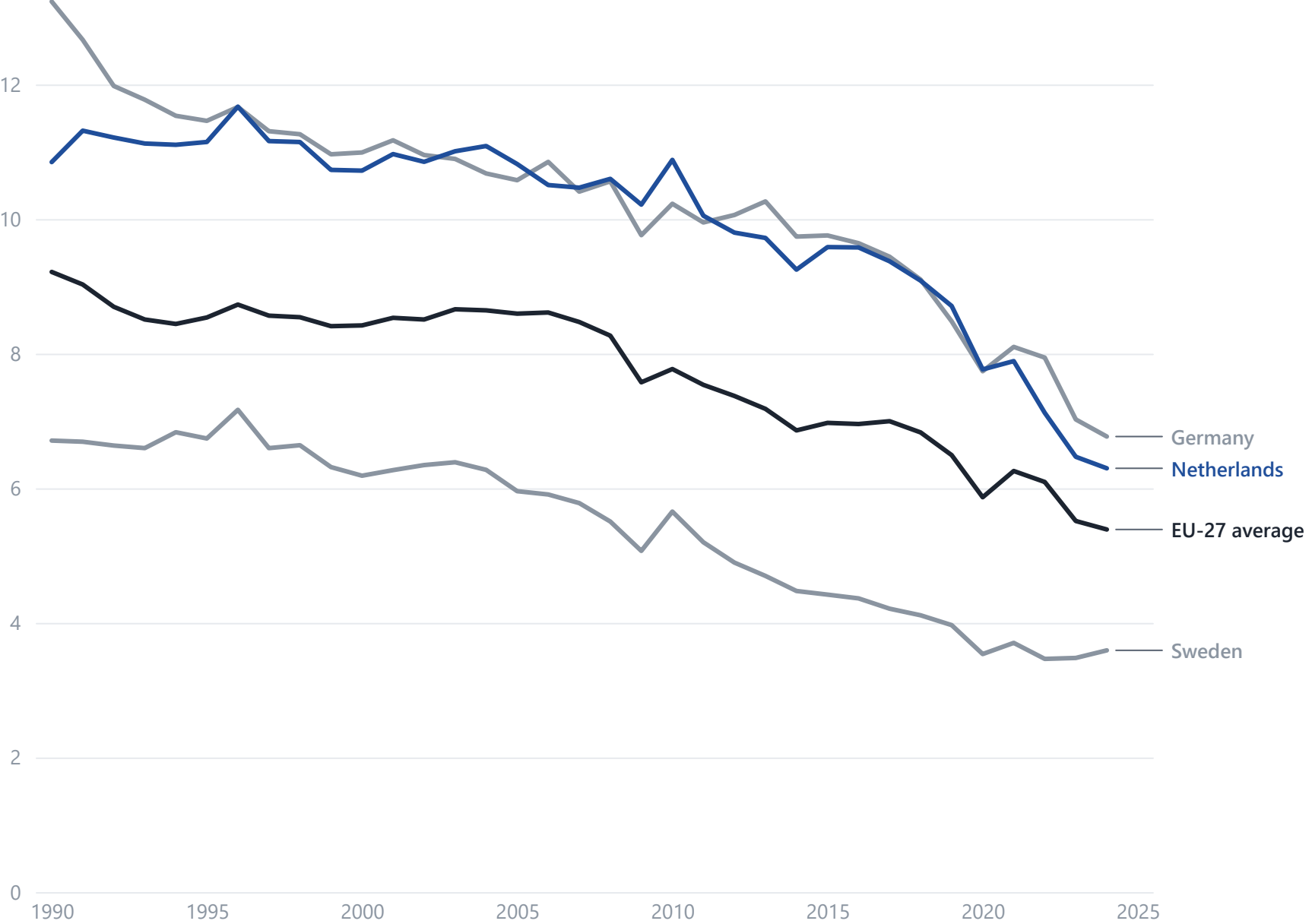
CO₂ per person by country, 2024

Tonnes per person. Netherlands in blue, EU-27 average in black.



CO₂ per person over time, 1990–2024

Tonnes per person. Netherlands and the EU-27 average, with Germany and Sweden for comparison.

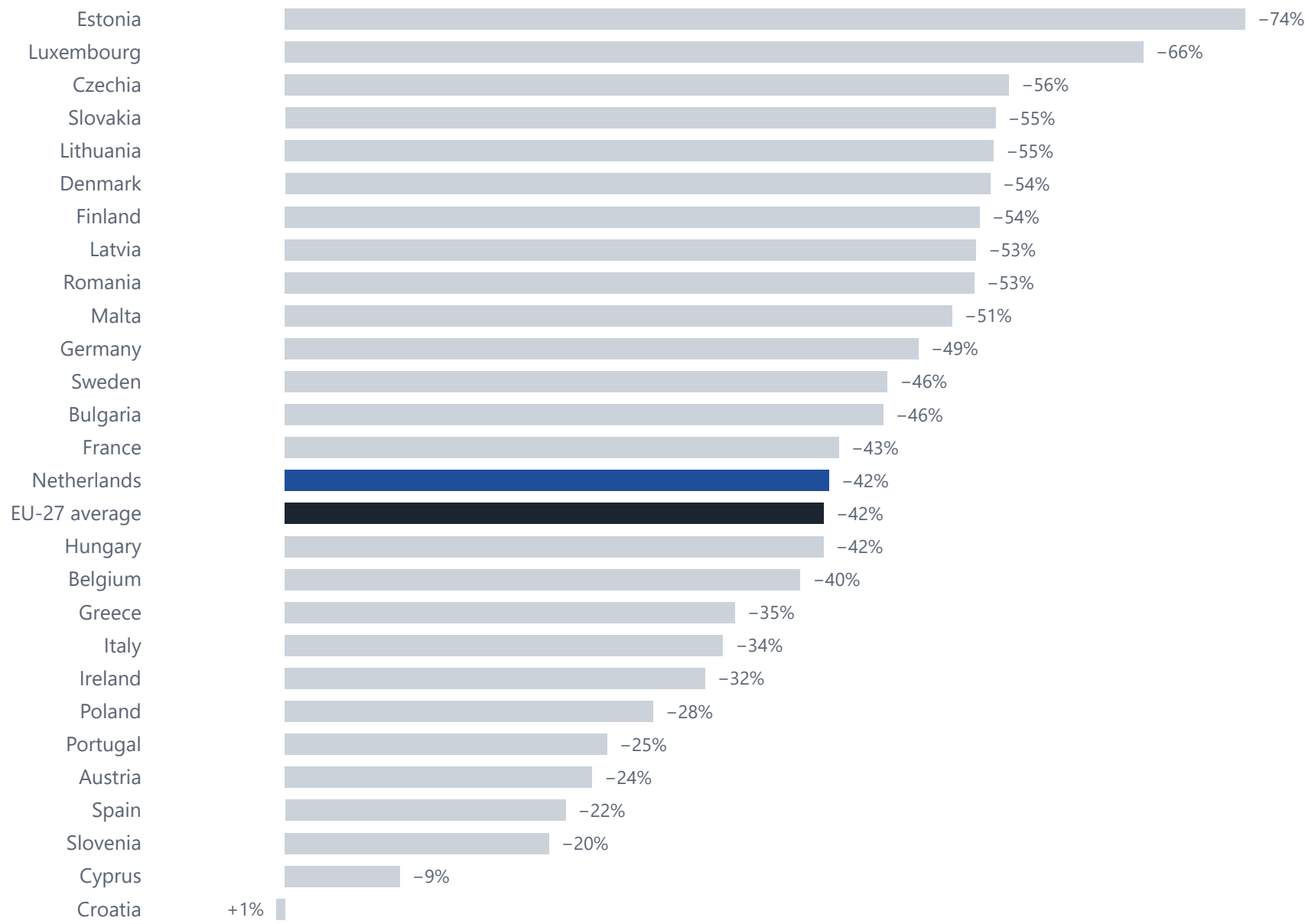


Progress since 1990 and outlook to 2030

Change in CO₂ emissions per person, 1990–2024, and the Dutch trend compared with an indicative 2030 benchmark.



Change in CO₂ per person, 1990–2024
Largest reduction at the top. Netherlands in blue, EU-27 average in black.



Netherlands: CO₂ per person and an indicative 2030 benchmark
Tonnes per person. Dashed line: 55% below 1990 (4.88 t). Indicative only, as the official target covers all greenhouse gases.



Conclusions

Answers to the sub-questions.

SQ1

How did Dutch emissions change?

–42% since 1990: from 10.85 to 6.30 tonnes per person.
1990–2010: almost no change (+0.3%), while the EU fell 16%.
After 2010: a fast decline (–42%), faster than the EU (–31%).

SQ3

Who reduced the most?

Estonia –74%, Luxembourg –66%, Czechia –56%.
The Netherlands is 15th of 27: about the same as the EU average (both –42%).
Countries that started high tended to reduce more ($r = -0.65$).

SQ2

Where does the Netherlands rank?

7th highest of 27 countries in 2024 (8th in 1990, 5th in 2010).
Still above the EU-27 average: 6.30 vs 5.39 tonnes.
Sweden (3.59) and France (3.97) emit about 40% less per person.

SQ4

Where is it heading by 2030?

If the recent trend continues: about 4.4–5.0 tonnes per person.
That is close to the indicative 4.88 t line (–55% vs 1990).
But the official target covers all greenhouse gases: PBL expects a 47–54% cut by 2030 (< 5% chance).

Answer to the research question

The Netherlands cut its CO₂ emissions per person at the same pace as the EU, but it started late and from a high level. It is still one of the higher emitters in the EU, so the fast progress since 2010 has to continue – and speed up – to reach the 2030 goals.

Limitations

- CO₂ only (no methane or nitrous oxide from agriculture). All greenhouse gases together fell only 36% by 2025 (CBS, 2026).
- Values per person, not totals. The population grew from about 15 to 18 million (CBS), so a 55% cut in total means about 4 t per person, not 4.88 t.
- Territorial emissions (imports not included)
- Trends assume the past continues